

Package ‘USFarmSafetyNetLab’

March 7, 2026

Type Package

Title US Farm Safety Net Lab

Version 0.0.0.9000

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Description This repository centralizes research outputs, analytical tools, and resources for exploring and evaluating the United States agricultural safety net programs. It supports analysis of key programs including the Federal Crop Insurance Program (FCIP), the Noninsured Crop Disaster Assistance Program (NAP), Price Loss Coverage (PLC), Agricultural Risk Coverage (ARC), and various ad-hoc disaster assistance programs.

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URL <https://github.com/ftsiboe/USFarmSafetyNetLab>

BugReports <https://github.com/ftsiboe/USFarmSafetyNetLab/issues>

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

VignetteBuilder knitr

Depends R (>= 4.1.0)

Imports readr, data.table, doBy, janitor, plm, sp, spdep, stringr, terra, tigris, xml2

Remotes github::dylan-turner25/rfcip, github::UrbanInstitute/urbnmapr, github::dylan-turner25/rfsa

Suggests dplyr, rvest, knitr, rmarkdown, rfcip, withr, piggyback, testthat (>= 3.0.0)

LazyData true

LazyDataCompression gzip

Cite-us If you find it useful, please consider starring the repository and citing the following studies

- Tsiboe, F. and Turner, D. (2025). ``Incorporating buy-up price loss coverage into the United States farm safety net." Applied Economic Perspectives and Policy.
- Tsiboe, F., et al. (2025). ``Risk reduction impacts of crop insurance in the United States." Applied Economic Perspectives and Policy.
- Gaku, S. and Tsiboe, F. (2024). Evaluation of alternative farm safety net program combination strategies. Agricultural Finance Review.

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calculate_mode	<i>Calculate the Statistical Mode</i>
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Description

Returns the element that occurs most frequently in a vector.

Usage

```
calculate_mode(x, na.rm = TRUE)
```

Arguments

x	A vector of any atomic type (numeric, character, factor,).
na.rm	Logical; should missing values be ignored? Defaults to TRUE. If FALSE and x contains any NAs, the function returns NA.

Details

Internally the function:

1. Optionally removes NAs (`na.rm = TRUE`).
2. Builds a lookup table of unique values via `unique(x)`.
3. Counts the frequency of each unique value with `tabulate(match(x, ux))`.
4. Returns the value with the maximum count.

Because it relies on base R functions, the implementation is vectorised and generally fast for typical data-frame column sizes.

Value

A single value giving the modal element of x. If two or more values are tied for the highest frequency, the first one encountered in x is returned.

Description

Initializes the working environment for a project by creating required directories, setting useful global options, and fixing the random seed.

Usage

```
setup_environment(
  year_beg = 2001,
  year_end = as.numeric(format(Sys.Date(), "%Y")),
  seed = 1980632,
  project_name,
  local_directories = list(file.path("data-raw", "output"), file.path("data-raw",
    "scripts"), file.path("data")),
  fastscratch_root = NULL,
  fastscratch_directories = NULL
)
```

Arguments

year_beg	Integer. Beginning year of the analysis (default: 2001).
year_end	Integer. Ending year of the analysis (default: current system year).
seed	Integer. Random seed for reproducibility (default: 1980632).
project_name	Character. Project name (required). Used to build fast-scratch directory paths.
local_directories	List of project-local directories to create (default: list("data-raw/output", "data-raw/scripts", "data")).
fastscratch_root	Optional character. Root directory for fast-scratch files. If NULL, it is set automatically: • Windows: "C:/fastscratch" • Linux/macOS: "/fastscratch/<username>"
fastscratch_directories	List of fast-scratch subdirectories (relative to <fastscratch_root>/<project_name>) to create. If NULL, no fast-scratch subdirectories are created and wd is returned as an empty list.

Details

The function ensures the requested directories exist, creating them if necessary. Directory keys in the returned wd list are the basenames of the provided fastscratch_directories.

It also sets the following options:

- options(scipen = 999) (turns off scientific notation)
- options(future.globals.maxSize = 8 * 1024^3) (~8 GiB)
- options(dplyr.summarise.inform = FALSE) (quiet **dplyr**)

Finally, the random number generator is seeded with the provided seed.

Value

A list with:

wd Named list of created fast-scratch directories. Empty if fastscratch_directories = NULL.

year_beg Starting year (integer).

year_end Ending year (integer).

seed Seed value used for RNG.

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